

Damage-triggered hydraulic isolation in a multifunctional biomimetic capillary skin

A falsifiable computational design study

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PHYSICAL VALIDATION REQUIRED**

All numerical results in this document are generated by explicit computational models and assumptions. They must not be represented as measurements, proof of physical feasibility, or clinical/environmental performance. The document is designed to make the concept falsifiable before fabrication.

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Abstract

Biomimetic materials commonly reproduce individual natural functions, such as humidity-driven actuation, atmospheric-water capture, vascular transport or damage repair. A more demanding objective is to integrate multiple functions through a common physical state variable rather than by adding sensors, pumps and controllers. Here we propose Adaptive Capillary Skin (ACS), a distributed fluidic material in which water is intended to serve simultaneously as transported resource, thermal medium, hydraulic state carrier and trigger for local damage isolation. The central hypothesis is that a puncture-induced pressure/flow perturbation can recruit fluid to a confined swelling collar around the damaged branch, reduce leak conductance, and preserve transport through redundant bypass paths without centralized control. We formulate a low-order 5 x 5 capillary-network model using Hagen-Poiseuille conductances, an explicit phenomenological swelling/isolation law, and an idealized electrokinetic pressure-to-voltage surrogate. A preregistered computational gate requires $\geq 80\%$ leak reduction within 600 s, $\geq 70\%$ useful-flow retention, and a model sensing effect ≥ 5 assumed-noise standard deviations. Under broad parameter distributions, only 273/1000 Monte Carlo trials satisfy all gates, exposing sensitivity to swelling kinetics and residual aperture. A refined design envelope derived from those failures produces 997/1000 passing virtual trials; the nominal simulation predicts 92.8% late leak reduction, 97.4% useful-flow retention and 276 s to 80% isolation. These results are not experimental validation. They define material-response targets and a falsification protocol for a transparent 100 x 100 mm prototype. The study illustrates a route from biomimetic inspiration to a testable “material logic” in which damage itself provides the control signal for local hydraulic isolation.

Keywords: biomimetics; capillary networks; hygromorphic materials; self-healing materials; hydrovoltaics; damage tolerance; atmospheric water; falsification

1. Introduction

Many biological systems achieve apparently complex behavior without a centralized electronic controller. Plant organs bend through differential swelling; vascular networks distribute water and metabolites along redundant pathways; surface chemistry organizes droplet nucleation and transport; and local damage changes gradients that recruit fluids or repair processes. Biomimetic engineering has successfully translated many of these mechanisms into isolated technical functions. Hygromorphic bilayers can respond autonomously to ambient relative humidity, including at architectural scale [2,3]. Patterned wettability can enhance fog collection and directional drainage [6]. Hydrovoltaic systems can transform water motion or evaporation-associated ion transport into electrical signals or power [4]. Engineered microvascular materials can distribute healing agents or heat-transfer fluids and improve damage tolerance [7-9].

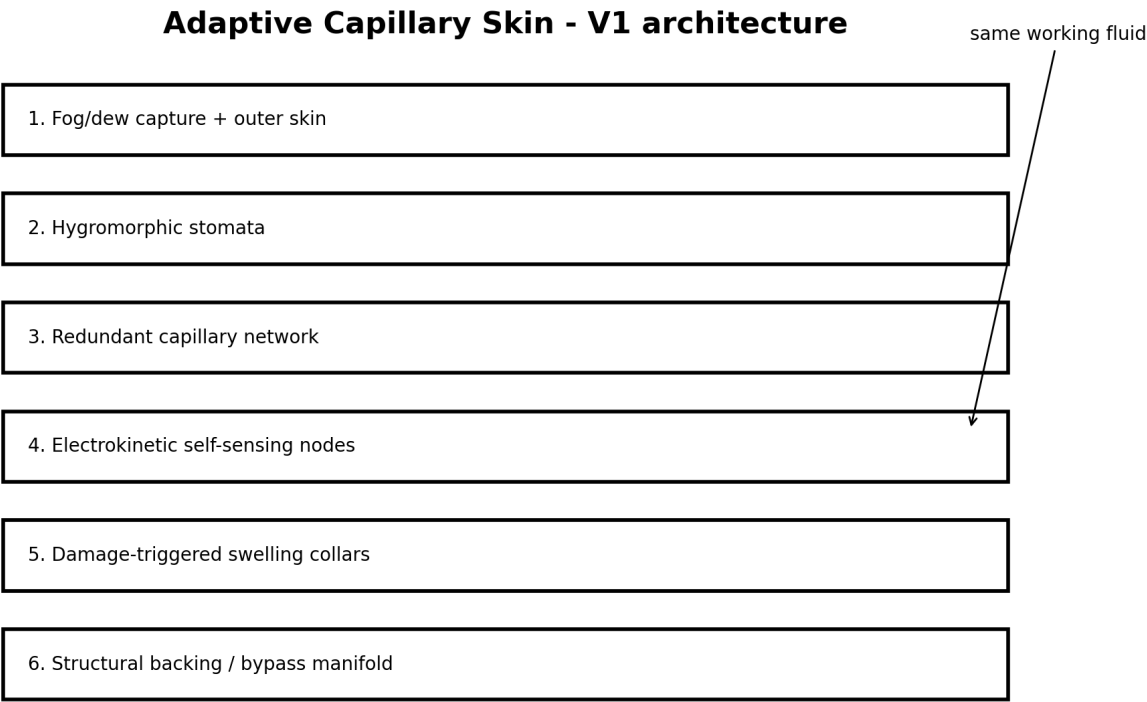
Integration, however, is often achieved by stacking subsystems: a collector feeds a reservoir, a pump drives a channel, a sensor reports state, a controller detects failure, and an actuator closes a valve. The result may work but can lose the material economy that made the biological model attractive. The

present study asks whether part of that control hierarchy can instead be embedded in the physics of a common working fluid.

The concept was motivated by a broader observation also seen in recent biomimetic fluidics: surface-energy release can perform useful transport without a conventional pump. Film-rupture actuated capillary enrichment, for example, uses a biomimetic liquid-film instability to move and concentrate airborne samples [1]. ACS does not attempt to reproduce that sampler. Rather, it applies a different design question to a vascular material: can a damage-created hydraulic perturbation itself actuate local isolation, while the surrounding network preserves useful flow?

We therefore introduce Adaptive Capillary Skin (ACS), a proposed layered material whose long-term vision combines (i) environmental water capture, (ii) humidity-responsive access, (iii) redundant capillary transport, (iv) evaporative thermal regulation, (v) electrokinetic state sensing and (vi) damage-triggered local hydraulic isolation. The present paper intentionally tests only the core hydraulic/sensing logic at low order and treats auxiliary functions as bounded design modules. This conservative scope is important: the individual technologies all have prior art, and no claim of experimentally demonstrated multifunctionality is made.

2. Design hypothesis and architecture



Core hypothesis: damage creates the hydraulic perturbation that drives its own local isolation.

Figure 1. Proposed ACS architecture. The candidate systems contribution is the coupling of functions around one capillary working-fluid network, especially damage-triggered local isolation without centralized control.

2.1 Material logic

The central causal chain is: environmental or stored water enters a redundant microvascular network; a puncture creates a low-resistance leak and perturbs local pressure; increased wetting of a confined swelling element reduces the effective leak conductance; flow redistributes through parallel paths; and a local electrokinetic readout may encode the changed pressure/flow state. Unlike a conventional valve, the isolation element is not commanded by a processor. The damage event is both disturbance and trigger.

2.2 Separation of core and auxiliary hypotheses

Table 1. Scope of modeled functions.

Module	Status in this paper	Why separated
Redundant hydraulics + damage leak	Explicitly modeled	Core question
Swelling-collar isolation	Phenomenological dynamic model	Core question; material law to validate experimentally
Electrokinetic self-sensing	Idealized surrogate	Tests detectability concept, not actual power/performance
Hygromorphic gating	Parametric hysteresis model	Independent coupon test required
Fog/dew capture	Bounding calculation only	Depends strongly on environment and geometry
Evaporative cooling	Psychrometric/energy bound only	Requires measured evaporation and heat-transfer coefficients

3. Methods

3.1 Hydraulic network

A 5 x 5 orthogonal node network was used to provide a minimal redundant topology. Circular channels of length 20 mm were assigned conductance $G = \pi r^4 / (8 \mu L)$. Mean radius was 90 μm in the nominal case with stochastic geometric variation. The left boundary was prescribed at 1000 Pa and the right at 0 Pa. Internal nodal pressures were obtained by linear mass conservation. This steady-state-per-time-step abstraction omits transient meniscus motion and fluid-structure coupling, allowing the study to isolate topological redundancy and valve kinetics.

$$G_{ij} = \pi r_{ij}^4 / (8 \mu L_{ij}) \quad (1)$$

$$Q_{ij} = G_{ij} (P_i - P_j) \quad (2)$$

3.2 Damage and isolation kinetics

Damage was introduced at the central node at $t = 300$ s as a shunt conductance to ambient pressure. A swelling collar was represented by a 5 s wetting delay followed by exponential decay in leak conductance from G_0 to $G_{\min}$ with time constant τ . The nominal parameters were $G_0 = 8 \times 10^{-12} \text{ m}^3 \text{ Pa}^{-1} \text{ s}^{-1}$, $G_{\min}/G_0 = 0.02$ and $\tau = 90$ s.

$$G_I(t) = G_{\min} + (G_0 - G_{\min}) \exp[-(t - t_d - t_{\text{delay}})/\tau] \quad (3)$$

3.3 Idealized self-sensing surrogate

A local pressure difference spanning the damage region was mapped to an idealized streaming potential using $\Delta V = (\epsilon \zeta / (\mu \sigma)) \Delta P$. Nominal ζ potential was -30 mV, conductivity 0.01 S m⁻¹ and readout noise standard deviation 2 μ V. The purpose of this term is not to predict a practical generator; it is to determine whether a damage-associated pressure signal survives parameter variation strongly enough to justify an electrode experiment. The assumed readout noise is therefore reported as an input, and model “SNR” is not empirical sensor performance.

3.4 Falsification gates

Before interpreting the Monte Carlo ensembles, four pass criteria were fixed: late leak reduction $\geq 80\%$ relative to early peak leak; time to achieve 80% leak reduction ≤ 600 s after puncture; late useful outlet flow $\geq 70\%$ of undamaged baseline; and idealized self-sensing effect ≥ 5 times the assumed readout-noise standard deviation. A trial passed only if all criteria were met.

3.5 Monte Carlo stress tests

Two ensembles of 1,000 trials were evaluated. The broad-prior ensemble intentionally allowed slower collar kinetics, larger late residual openings and wider geometric/pressure/electrochemical variation. The refined design envelope was not chosen before the broad ensemble; it was derived from the observed failure modes and therefore represents a specification study rather than an independent validation set. That distinction is critical: the 99.7% refined pass rate cannot be interpreted as unbiased evidence of reliability.

Table 2. Monte Carlo parameter envelopes.

Variable	Broad prior	Refined envelope
Hydrogel tau	Lognormal around 120 s, broad spread	Lognormal around 90 s, clipped 45-160 s
Residual leak ratio	Broad 0.002-0.25	Approx. 0.003-0.05
Channel radius	65-115 μ m clipped	75-105 μ m clipped
Source pressure	700-1300 Pa	800-1200 Pa
Zeta potential	Around -30 mV, broader spread	Around -30 mV, narrower spread
Conductivity	0.005-0.020 S m ⁻¹	0.007-0.014 S m ⁻¹

3.6 Auxiliary design bounds

A hygromorphic gate was represented only for design visualization by two logistic branches centered at 65% RH on increasing humidity and 55% RH on decreasing humidity. Evaporative cooling was bounded using latent heat for 1 g h⁻¹ water evaporation from a 0.01 m² panel and capped by a Stull wet-bulb estimate [12]. Fog capture was bounded as LWC x air velocity x projected area x collection efficiency, using liquid-water content 0.1-0.5 g m⁻³, airspeed 0.5-2 m s⁻¹ and assumed efficiency 0.2-0.4. These values are scenarios, not measured ACS performance.

3.7 Reproducibility

The release includes the full nominal time series, both Monte Carlo datasets, the falsification parameter grid, auxiliary prediction tables and executable source. Random seeds are stored in the Monte Carlo

tables. The code and data are sufficient to reproduce the stated computational results without access to proprietary software.

4. Results

4.1 Nominal puncture response

Before $t = 300$ s the network settles to a steady baseline. The puncture produces an immediate leak and depressurizes the local region. As the modeled collar swells, leak conductance decreases and useful outlet flow recovers. At late time, leak reduction is 92.8%, useful-flow retention is 97.4%, and the 80% leak-reduction threshold is crossed 276 s after puncture.

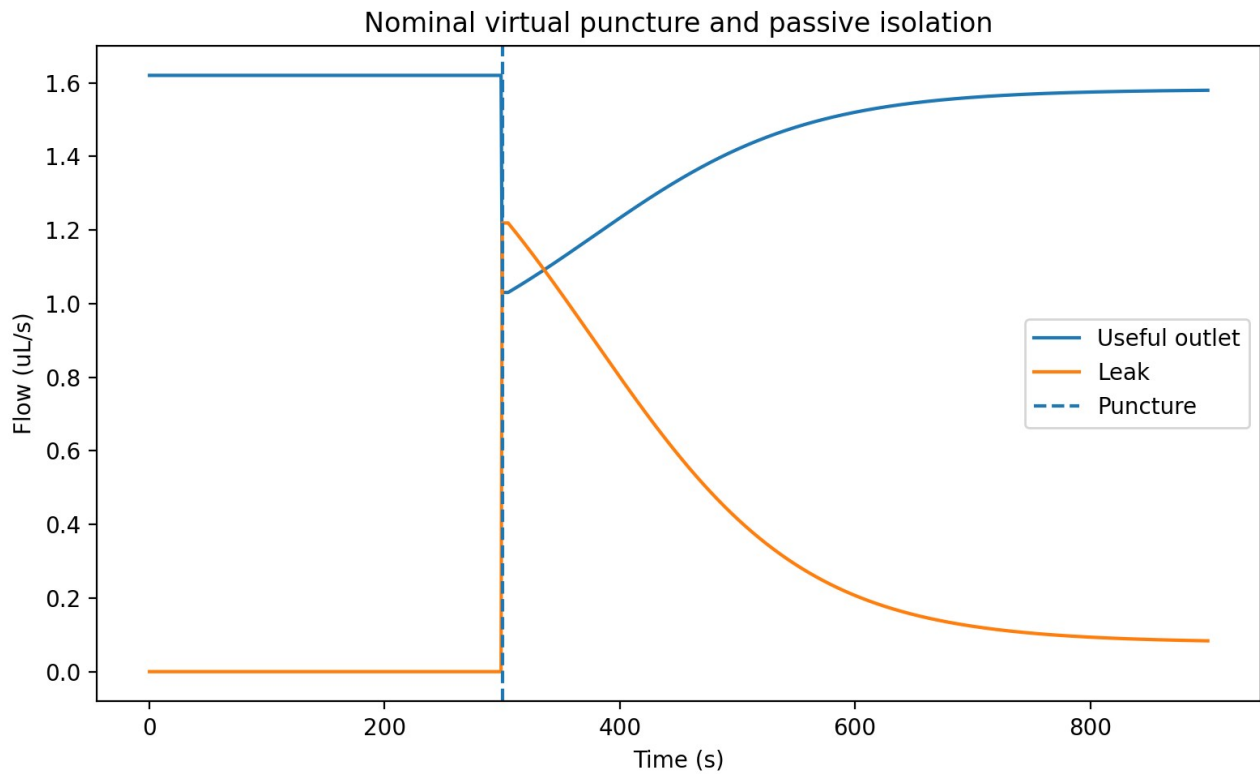


Figure 2. Nominal hydraulic response. Simulation only.

4.2 Damage-associated electrical-state surrogate

The local pressure perturbation produces a pronounced change in the idealized electrokinetic signal (Fig. 3). Because both the conversion law and readout noise are assumed, the large nominal virtual SNR is interpreted only as evidence that a pressure-state experiment is worth attempting. A real device could have much lower signal, greater source impedance and electrochemical drift.

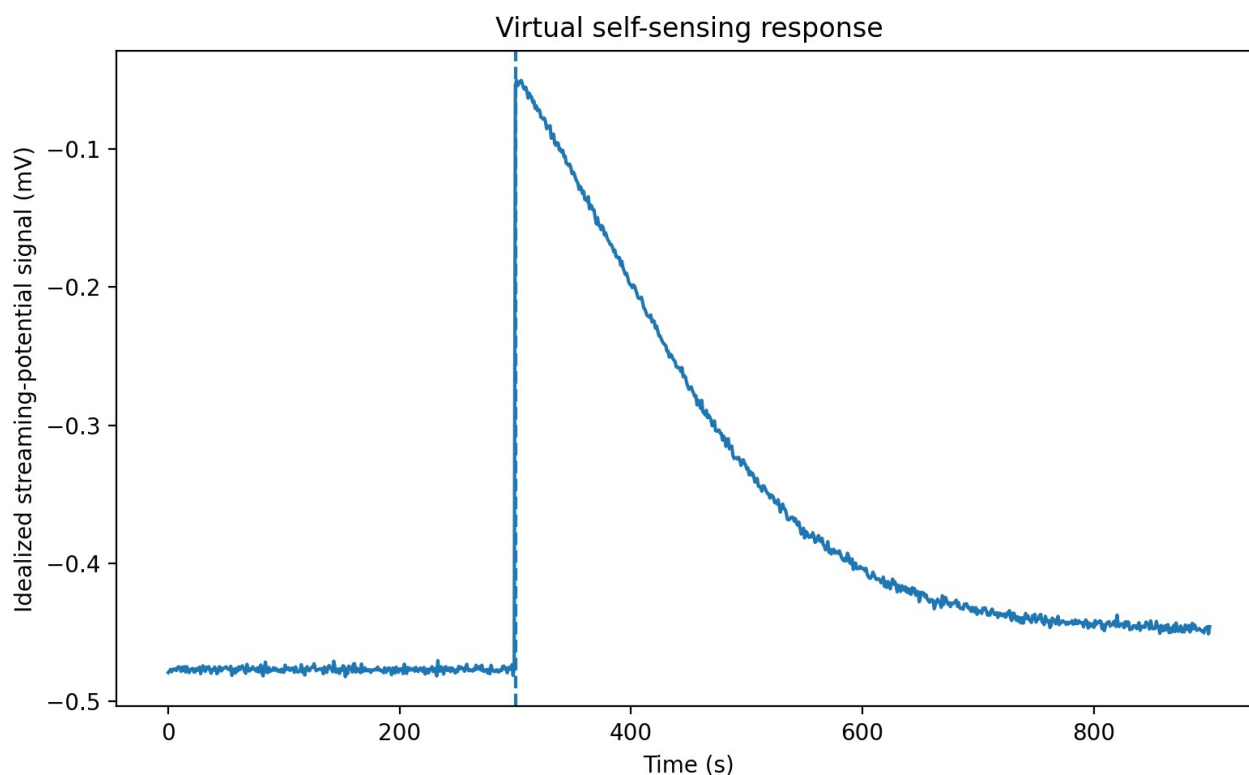


Figure 3. Idealized electrokinetic state signal. The curve is not measured voltage and does not establish harvestable power.

4.3 Broad-prior ensemble falsifies naive robustness

Only 273 of 1,000 broad-prior trials met every gate. This rejects a naive claim that any swelling element attached to any redundant microchannel network would provide robust self-isolation. Inspection of failures showed that late residual leak conductance and response time dominate the isolation gate, while geometric fourth-power sensitivity can alter the distribution of useful flow.

4.4 Refined envelope defines a material target

When the design is restricted to the response envelope identified from the broad failures, 997 of 1,000 trials pass all computational gates. The result should be read as a material-property target: a real swelling collar must close to roughly a few percent of the initial leak conductance and must do so on the order of minutes, not as proof that a particular polymer already satisfies those requirements.

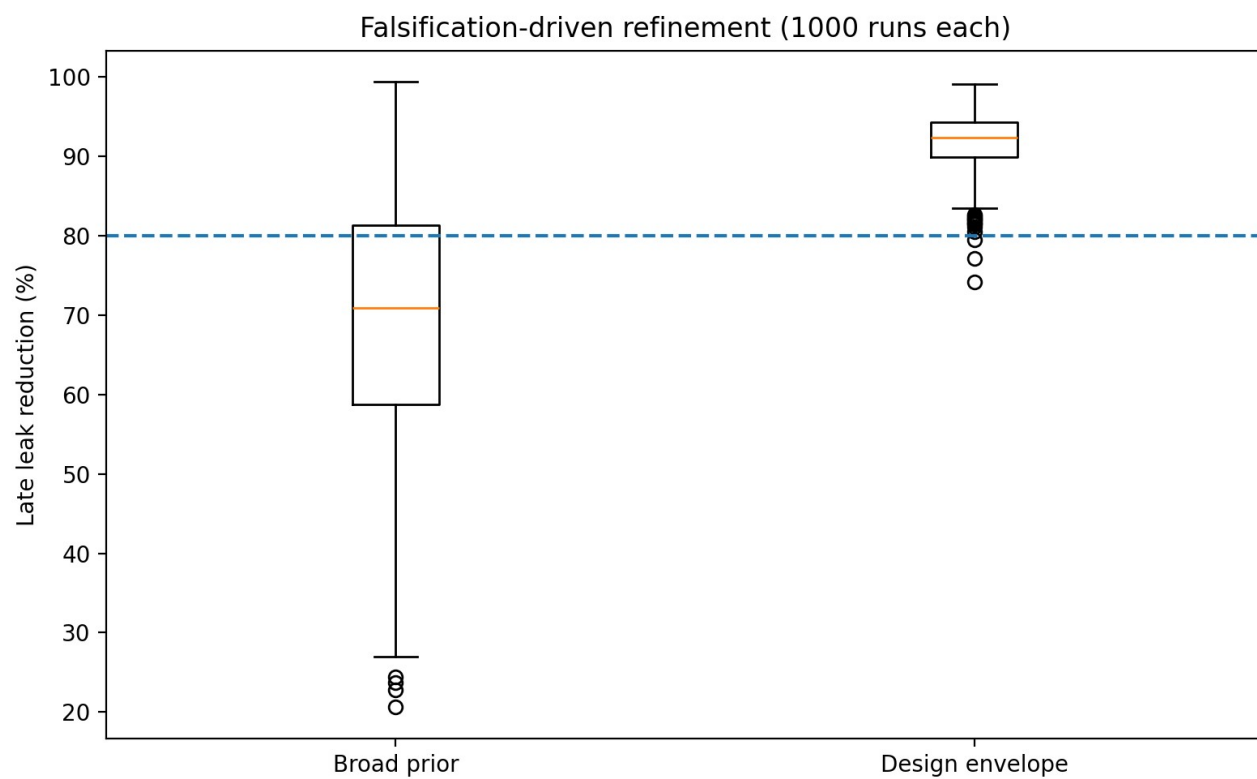


Figure 4. Leak-reduction distributions under broad and refined parameter envelopes. The refined ensemble is a design-specification study, not an independent validation cohort.

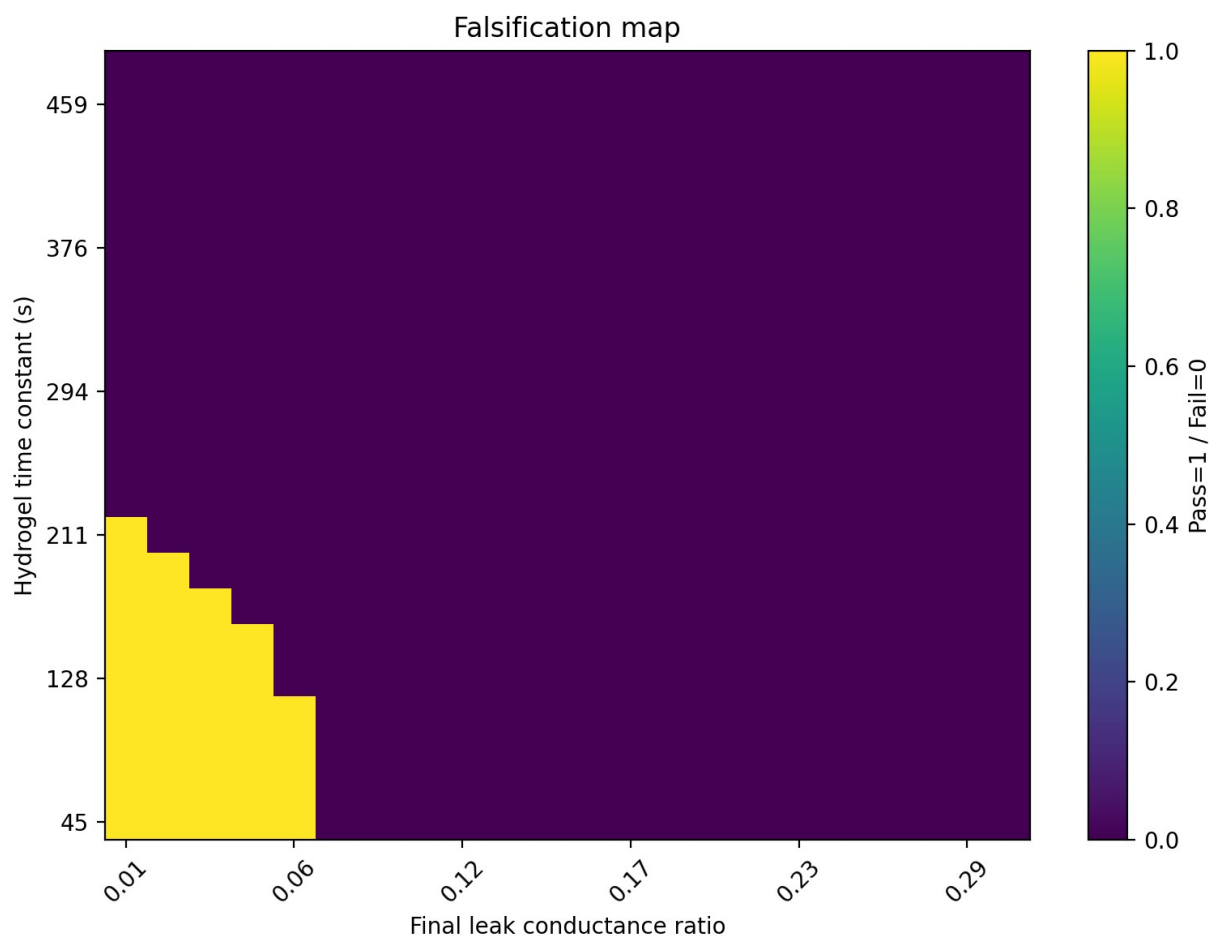


Figure 5. Pass/fail map across hydrogel response time and late residual leak conductance. The boundary defines candidate material targets for prototype screening.

4.5 Auxiliary functional predictions

The proposed humidity gate produces the expected hysteretic response by construction and therefore carries no evidential weight until a real coupon is characterized. Similarly, the simple cooling analysis shows only that at a 1 g h^{-1} evaporation rate the thermal effect would be bounded by latent heat and psychrometric wet-bulb conditions. These modules are retained to constrain future experiments, not to inflate the present result into a complete multifunctional demonstration.

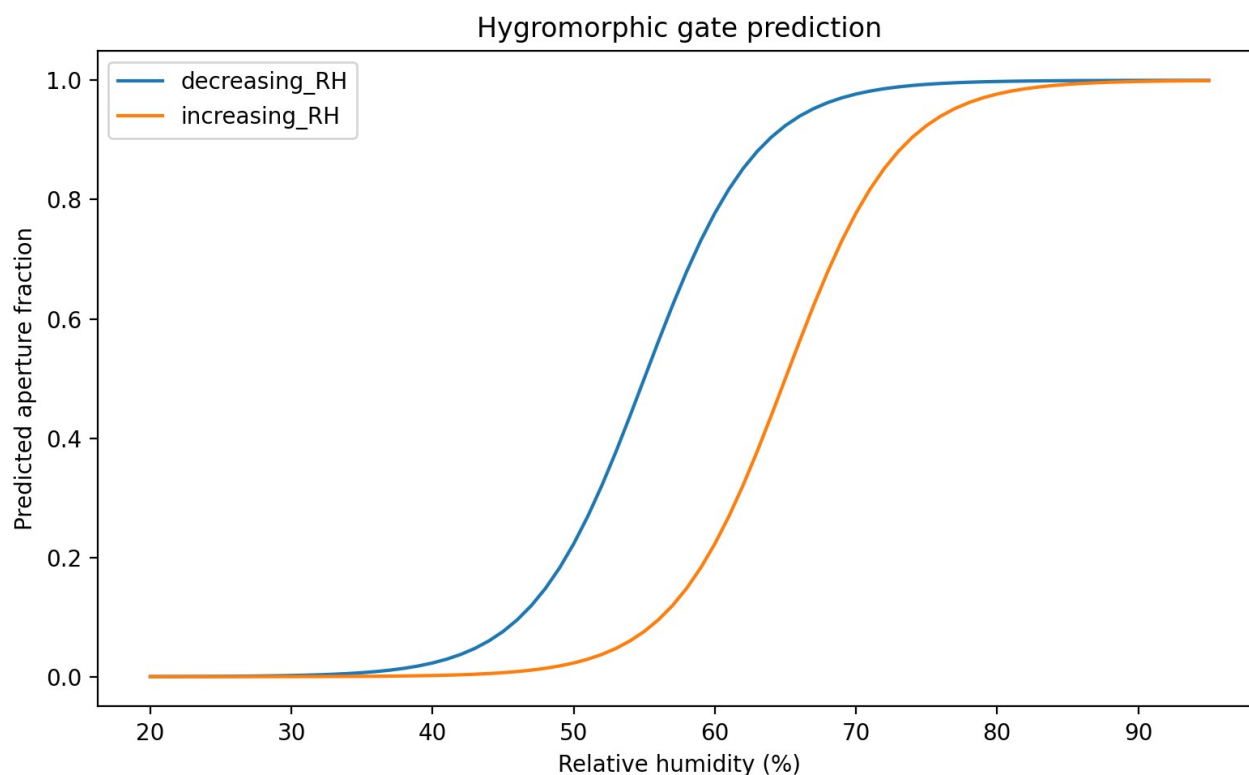


Figure 6. Parametric hygromorphic-gate design curve. Simulation only.

5. Discussion

5.1 From biomimetic components to embedded control

The most interesting property of ACS is not the number of proposed functions. Each function has precedents. The design question is whether the working fluid can become a shared physical state variable that coordinates them. In the damage pathway, the system does not detect a puncture and then issue a command; the puncture changes hydraulic resistance, which changes local wetting, which changes the geometry of the leak path. This causal compression is analogous to material-based actuation in hygromorphs, where environmental moisture both senses and actuates the response [2,3].

5.2 Relation to microvascular self-healing

Microvascular self-healing materials have already shown that embedded networks can deliver healing agents and that vascular architectures can be integrated into structural materials [7-9]. ACS differs in emphasis: the V1 endpoint is not reconstruction of fracture surfaces by polymerization but rapid hydraulic self-isolation and preservation of network service. A later self-sealing film could be added, but the first experiment should avoid conflating closure, healing chemistry and structural recovery.

5.3 Relation to hydrovoltaic systems

Hydrovoltaic devices demonstrate that water flow and evaporation across charged materials can generate measurable electrical outputs [4], and integrated fog-harvesting/triboelectric systems have been reported [5]. ACS therefore does not claim water-to-electric transduction as a new phenomenon. Its proposed role is narrower: a local, possibly battery-free indication of hydraulic state. A successful

prototype would need to report source impedance and extractable power, not only open-circuit voltage, before any energy-harvesting claim could be justified.

5.4 Thermodynamic accounting

No component creates energy. Capillary transport is powered by interfacial free energy and pressure gradients; hygromorphic motion is powered by chemical potential changes associated with water sorption; evaporative cooling consumes latent heat; and any electrical output ultimately derives from the same environmental gradients and fluid motion. This accounting is important because a multifunctional passive material can otherwise be mistakenly described as self-powered in a thermodynamically ambiguous way. In ACS, “passive” means no external pump or active controller is required for the intended local response, not that energy is absent.

5.5 Why the broad ensemble matters

A central methodological result is the 27.3% broad-prior pass rate. If only the nominal run had been reported, the design would look almost automatically successful. The failed ensemble instead shows that the concept has a narrow enough physical requirement to be falsifiable. The refined 99.7% pass rate is useful only because it converts that failure into a specification: screen candidate collars for kinetics and residual aperture before attempting full integration.

5.6 Manufacturability and likely failure mechanisms

A 100 x 100 mm transparent demonstrator is feasible with common microfabrication routes, but several effects can invalidate the low-order model: trapped air can stop capillary priming; channel roughness and contact-angle hysteresis can introduce threshold behavior; hydrogels can swell into bypass paths or fatigue; salts can change swelling and zeta potential; microbial growth and particulate fouling can alter surface chemistry; and thin channels can lose fluid through evaporation. These are not secondary engineering details - they are candidate falsifiers of the integrated architecture.

6. Experimental validation plan

Table 3. Proposed physical falsification experiments.

Experiment	Control	Primary endpoint	Reject/modify ACS if...
Hydraulic puncture + collar	Same network without collar; sham puncture	Leak reduction at 600 s; useful-flow retention	<80% leak reduction or <70% useful flow across a preregistered specimen set
Collar kinetics coupon	Unconfined hydrogel and inert ring	Swelling ratio, closure force, response time	Required closure cannot be achieved within model envelope
Electrokinetic readout	Surface-neutral insert; sham; conductivity series	Damage-vs-control differential signal and source impedance	Signal is not repeatable beyond drift/noise or provides no useful information
Hygromorphic aperture	Inert geometry control	Aperture vs RH over repeated cycles	Response is too small/slow or degrades rapidly
Fog/dew capture	Unpatterned surface	Net mass capture per area	No advantage over control after blank correction

Experiment	Control	Primary endpoint	Reject/modify ACS if...
Evaporative cooling	Dry identical geometry	Temperature/heat-flux difference with measured evaporation	Thermal effect is not reproducible or requires impractical water throughput

The first paper following this computational study should publish all specimens and traces, including failed isolation events. If physical results require fitting tau or Gmin after the fact, the manuscript should explicitly distinguish calibration from prediction and should test the calibrated model on independent specimens.

7. Limitations

- The present study contains no fabricated prototype and no physical measurements.
- The dynamic hydrogel law is phenomenological and can substantially over-simplify diffusion, confinement, viscoelasticity and fatigue.
- The network model does not resolve multiphase priming, evaporation within channels, wall deformation or transient inertial effects.
- The electrokinetic model is a surrogate and cannot establish practical electrical output or sensing reliability.
- The atmospheric-water and cooling modules are only order-of-magnitude design bounds.
- The refined Monte Carlo ensemble was derived from broad-ensemble failures and is therefore not independent validation.
- Novelty and freedom to operate have not been legally established; prior art exists for every major component class.

8. Conclusion

This computational study reframes a multifunctional biomimetic surface as a falsifiable hydraulic-control problem. A low-order redundant network does not self-isolate robustly under broad assumptions; only 27.3% of broad Monte Carlo trials meet all gates. The failures define a concrete target for the swelling element. Conditional on a response time and residual leak aperture within the refined envelope, 99.7% of virtual trials pass, and the nominal model preserves 97.4% useful flow while reducing the leak by 92.8%. These values are design predictions, not performance claims. The next decisive step is a transparent hydraulic prototype that can fail in public, with calibrated damage, interchangeable collars, raw flow traces and matched controls. If that experiment succeeds, ACS would provide a physically compact example of damage acting not only as a disturbance but also as the signal that initiates its own local isolation.

Data availability

All computational data underlying this manuscript are supplied with the accompanying open research package as CSV files and an XLSX workbook. No physical experimental datasets are reported.

Code availability

The executable low-order model is supplied as `acs_simulation_model.py`. The code uses Python, NumPy and pandas and stores random seeds in the Monte Carlo output tables.

Author contributions

S.K.: conceptualization, project direction, interpretation and manuscript responsibility. AI-assisted implementation and drafting are disclosed below. The contribution statement should be revised if additional human collaborators join the project.

Competing interests

The author declares no competing interests at the time of this draft. This statement must be rechecked before submission.

AI-assistance disclosure

OpenAI ChatGPT (GPT-5.6 Sol) assisted with literature orientation, numerical implementation, visualization and manuscript drafting. The human author reviewed the concept and is responsible for source verification, scientific interpretation, corrections, submission decisions and any future experimental claims. The AI system is not an author.

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